Special Session II: Presentation 4

Emotion Recognition in Humorous Human-Computer Communication: A Conversational System Using Humor According to User Emotions

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The issues of emotions, humor and their role in communication continue to gain the attention of researchers in both cognitive and computer science. This work connects these areas. We first analyzed how humor and emotions work in human-human interactions and on this basis constructed a conversational system for Japanese which recognizes user emotions and use humor in response to them.

The components of the system are as below.

1) **The conversational system** – talks with users in a non-task-oriented manner. It uses Internet to generate adequate responses.

2) **The emotion-from-text recognition system** - detects emotions from the textual layer of conversations in Japanese. It uses a lexicon of emotive expressions, based on Nakamura's emotions classifications, modified for our needs. The system recognizes emotive expressions and associates them with particular emotive states. It also uses Internet to extract emotive associations toward inputted phrases.

This textual-layer-analysis approach is based on the "language-as-context" theory which states that in most cases of emotional cognition the linguistic layer gives us important contextual information about interlocutor's emotion. Thus, the setup used in this system is based on how humans recognize emotions.

3) **The humor generator** - uses the Internet and a database of sound transformation patterns to generate Japanese puns according to user utterances.

The sound transformation patterns are based on our previous research, in which we analyzed human-created Japanese puns and mora transformations that occur in their construction. The results were used in our humor generator, which means that phonetically it was based on how humans create puns.

Above systems were merged into a humor-and-emotion-equipped conversational system. Basing on existing works on humor and emotions, we assumed that in human-computer

interaction, the latter could use humor to make the interlocutor feel better. In order to do that, first the system has to detect the human partner's emotions. This function in our joking system is performed by the emotion recognition module. Based on the outcome of this analysis, the system decides if to tell a joke. The decision patterns are: 1) if the user's emotive state is negative/neutral: tell a joke; 2) otherwise: do not tell a joke.

The system was evaluated in two experiments: user-oriented and automatic (emotiveness-analysis-based).

In the first experiment, human evaluators were asked to compare two systems: with and without the emotion-humor module. The results showed that the former system was evaluated as better, more interesting and making users feel better than the one without humor.

In the second experiment, the chat logs acquired in the first experiment were analyzed with the emotion recognition system. The results showed that the system with humor elicited more emotions in users, most of which were positive or changed to positive during the interactions.

These results show that our system, although not perfect, can make users feel better. In the future, we plan to investigate which phonetic patterns in jokes trigger which particular emotions.

keywords: emotions, emotion recognition, humor generation, human-computer interaction, conversational systems